The University of Manchester

EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Week 2 Practical - Introduction to Differentiation (Solutions)

Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Describe the concept of differentiation and explain its significance in calculus.

(2) Identify and interpret common notation used in differentiation.

(3) Apply common laws of differentiation to solve theoretical and real-world problems.

Exercise 1: Rearranging expressions

- Differentiate the following functions with respect to x:
  (a) $y = 4x^3 + 2x^2 + 5x + 6$
  (b) $f(x) = \sqrt{x}$
  (c) $g(x) = 4x^4 + e^x - \cos(x)$
  (d) $h(x) = e^{5x+4}$

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

SOLUTION 1: REARRANGING EXPRESSIONS

- Differentiate the following functions with respect to x:
  (a) $\frac{dy}{dx} = 12x^2 + 4x + 5$
  (b) $\frac{df}{dx} = \frac{1}{2\sqrt{x}}$
  (c) $\frac{dg}{dx} = 16x^3 + e^x + \sin(x)$
  (d) $\frac{dh}{dx} = 5e^{5x+4}$

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

Exercise 2: Gas Leak

- Imagine we are studying the volume of helium gas in a tank, which is leaking, over time. If we say that the volume of water in the system can be described in time t as:

$$V(t) = V_0 - \frac{t}{60}$$

  where $V_0 = 1$ is the volume of gas (in m^3) and $0 < t < 60$ (in mins).
- What is the rate of the leak?
- Do you think the rate is sensible?

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
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SOLUTION 2: GAS LEAK

- To obtain the rate of the leak we differentiate with respect to time t:

$$\frac{dV}{dt} = -\frac{1}{60}$$

which is constant.

- It is unlikely that the rate of the gas leak is constant.

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
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Exercise 2: Gas Leak

- In reality the gas would leak at a non-constant rate. If we say that the volume of gas in the tank can be described as:

$$V(t) = V_0 \exp\left(-\frac{t}{6}\right)$$

where $V_0 = 1$ is the volume of gas (in m^3) and $0 < t < 60$ (in mins).

- What is the rate of the leak at 6 minutes?
- Compare this to a gradient calculated numerically using the central difference method with a time step of 0.1 minutes either side of 6 minutes.

Function $y = f(x)$	**Derivative $\frac{dy}{dx}$**
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
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Solution 2: Gas Leak

- To obtain the rate of the leak we differentiate with respect to time t:

$$\frac{dV}{dt} = -\frac{1}{6}\exp\left(-\frac{t}{6}\right)$$

- At $t = 6$ the (analytical) rate of the leak is:

$$\frac{dV}{dt}(6) = -\frac{1}{6}\exp\left(-1\right) = 0.061313$$

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x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
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Solution 2: Gas Leak

- At $t = 6$ the (analytical) rate of the leak is:

$$\begin{aligned}\frac{\Delta V}{\Delta t}(6) &= \frac{V(6.1) - V(5.9)}{6.1 - 5.9} \\ &= \frac{\exp(-6.1/6) - \exp(-5.9/6)}{0.2} \\ &= -0.061316\end{aligned}$$

- It is fairly accurate with a small step-size:

$$\begin{aligned}\text{Error } &= 100 \times \frac{-0.061313 + 0.061316}{-0.061313} \\ &= 0.004\%\end{aligned}$$

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